India STEM Lab

Empowering Innovation Through STEM Education

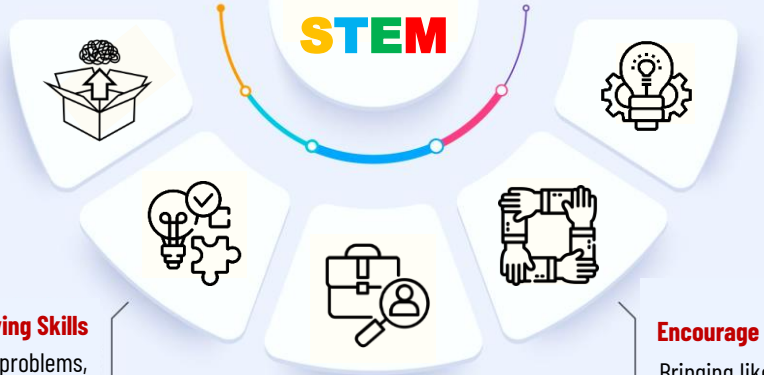
AICRA
All India Council For Robotics & Automation

Today, we are witnessing technology like AI (Artificial Intelligence), which used to be in science fiction, becoming an integral part of our lives. Robotics and drone technology have already knocked on our doors. Hence, we must break free from old thinking and embrace new horizons. We need to prepare our children for this. I wish to see our schools organizing interactive sessions related to future technologies.

Shri Narendra Modi
Prime Minister of India

SCIENCE | TECHNOLOGY | ENGINEERING | MATH

WHY STEM



Creative and Out-of-Box Thinking

Conceptualizing, applying, analysing, and evaluating information through observation

Encouraging Inventions in STEM Field

Edge over traditional education and creating open environment for inventions

Problem Solving Skills

Developing ability to identify problems, brainstorm and analyse answers and implement the best solutions

Encourage Team-Work and Collaboration

Bringing like-minded students together in order to efficiently and effectively complete projects

Making Youth Employable and Ready for Future Jobs

Access to advance technologies that provides youth not only with technical and vocational but also core employability skills

AICRA
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Introduces

India STEM LAB

1 GRANT TO SETUP STATE-OF-ART ISL

AICRA would provide ISL setup grant of INR 12.0L to school which is provisioned from corpus of Govt Fund + CSR + Pvt Investor.

2 ON-SITE TRAINING EXPERTS

AICRA certified training expert will be deployed at institute to train students and help them to develop advance projects.

3 MAINTENANCE & TECHNOLOGIES UPGRADE

Regular lab maintenance and technology upgrades in India STEM Lab ensure optimal functionality, safety, and access to cutting-edge tools for effective learning.

4 MODULES & PROJECTS UPGRADE

India STEM Lab emphasizes continuous improvement through updated modules and projects, enhancing learning experiences and keeping space with technological advancements and industry trends

5 ENTREPRENEURSHIP DEVELOPMENT

AICRA under IndiaFirst Startup Program, offers unique support system for students who wish to persue entrepreneurship goals.

6 INDUSTRY FOCUSED UNIQUE ISL ECOSYSTEM

The industry-focused STEM Lab ecosystem bridges education and real-world applications, fostering innovation, collaboration, and essential skills for future careers.

INR 12,00,000 GRANT FOR 2000 SCHOOLS

State-of-Art Infra and Design

ISL SETUP STRUCTURE

Projects that combine various STEM fields to solve real-world problems. These sections can provide a comprehensive and engaging experience for students interested in STEM!

Instrumentation Section
Learn measurement techniques and sensors to collect data for experiments and projects.

Mechanical Section
Focus on the principles of mechanics, simple machines, and mechanical systems

Electronics Section
Explore circuits, electronics, and basic electrical concepts using Arduino and Raspberry Pi.

IT and Software Section
Learn coding languages such as Python, Scratch, and Java, and develop software applications.

3D Printing Station
Create prototypes and models and other robotics projects from CAD designs.

UAV and Space Section
Explore Drone principles, design model aircraft, and study rocket science

Robotics Section
Design, build, and program robots for various challenges and competitions

Industry Automation Section
Automation technology, industrial robotics arm and its applications etc.

Gaming Section
Develop video games and interactive applications using game design principles

AI Section
Learn about machine learning, natural language processing, and AI applications

AR/VR Section
Use VR and AR technologies for immersive learning experiences

Display Zone
Students to showcase projects, innovations, inspiring creativity and collaboration.

Seamless Onboarding for Robotics Excellence

ISL INVESTMENT MODEL

AICRA is spearheading the India STEM Lab initiative in schools, creating a ₹200 crore corpus through government schemes/funding, CSR contributions, and private investments.
This project will enhance STEM education, fostering innovation and equipping students with essential skills for the future.

INSTITUTE OBLIGATIONS

- 200 – 250 Square Meter Space
- Dedicated SPOC (Single Point of Contact)
- Minimum commitment of 500 Students
- 5 to 10 PC with minimum i3 Core/8GB RAM
- Electricity Port and Internet Connection

INVESTMENT PLAN FOR ISL

Equipment & Components	INR 800,000
Furnishing & Branding	INR 200,000
IT Software & IP Cost	INR 150,000
Transportation and Setup	INR 50,000
Total Estimated Investment	INR 1200,000

AICRA have initiated the process of hiring technical training experts across India.

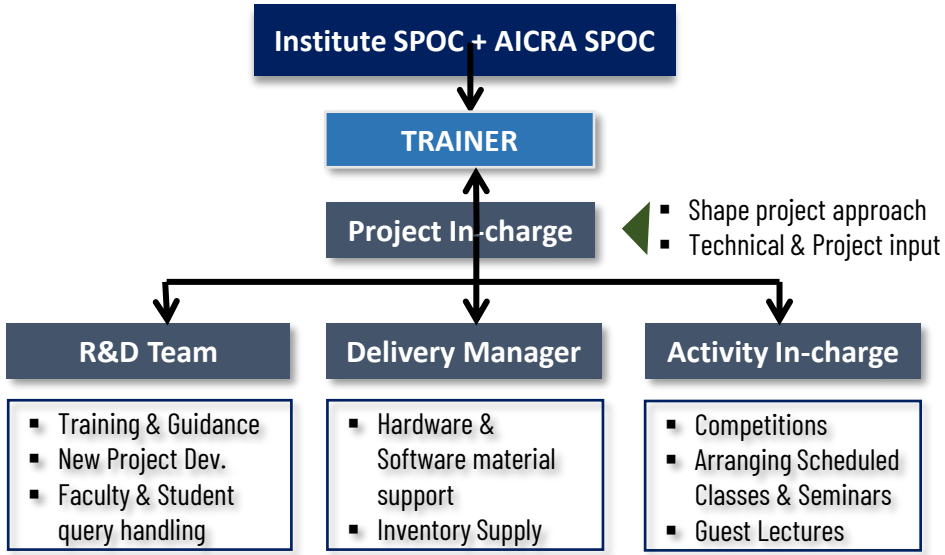
In 3 phases 2000 trainers will be hired and certified to become STEM Lab trainer.

- Qualification: B.Tech / MCA with 2 to 4 years of industry or academia experience
- Language: English/Hindi/ Regional Language
- Certification: Each trainer to undergo with AICRA's Trainer Development Program

Efficient Learning Execution

TEAM STRUCTURE

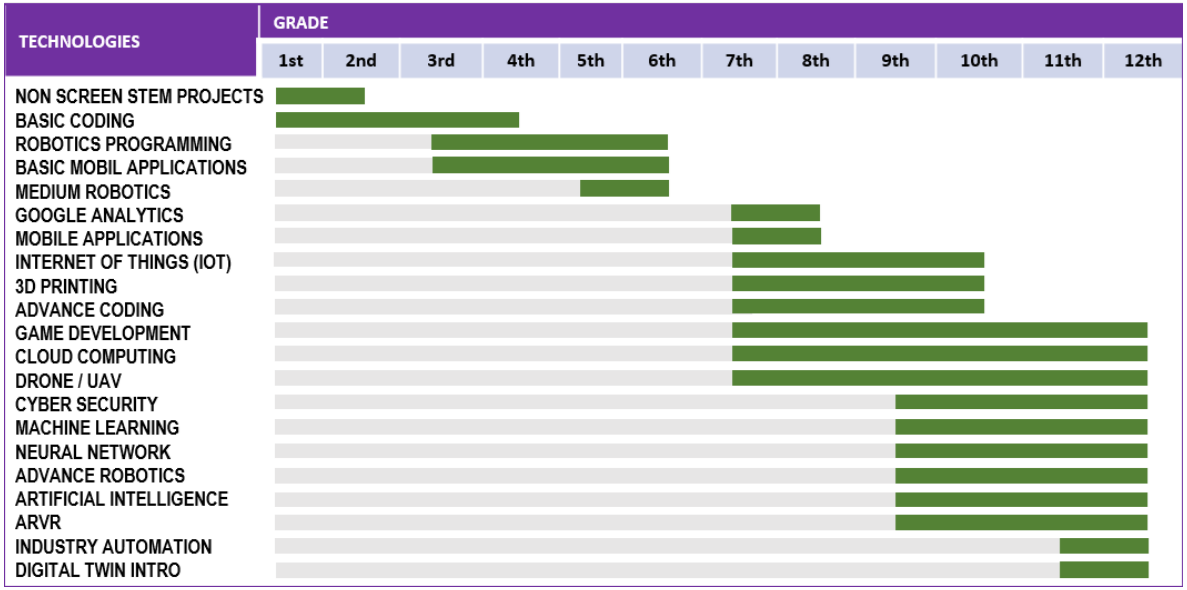
Empowering Innovation Through Robotics and eSports Excellence



- School may prefer daily/alternate/weekly training schedule for students. Deployment of the trainer would be as per schedule.
- Level 1 will be deployed at school and Level 2 Experts will also provide virtual and special sessions
- Rotation of trainers time to time, to ensure subject expert training

Explore, Create, Innovate: Future Awaits!

TECHNOLOGIES, STUDENTS TO LEARN IN ISL



Bridging Education & Industry

INDUSTRY FOCUSED UNIQUE ISL ECOSYSTEM

The India STEM Lab ecosystem focuses on industry collaboration, fostering real-world skills, innovation, and practical experiences for students' future careers.

BEYOND BASIC LEARNING

01 Tech Solutions to Real-Life Problems
Identifying regional/ day-to-day challenges and finding technology based solutions

02 Industry Focused Innovations
Innovations in major sectors like Agriculture, Healthcare, Aerospace etc.

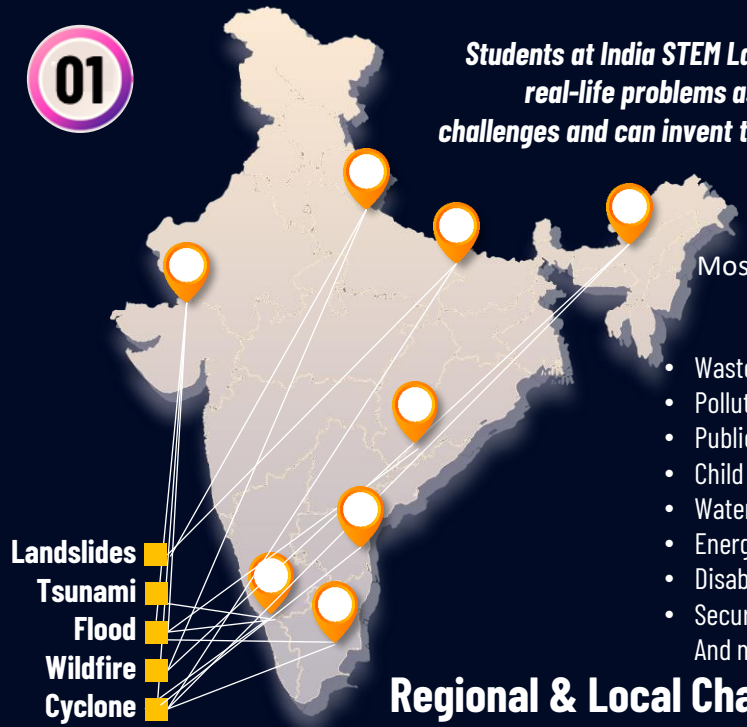
03 Industry Experts Guidance
Apart from regular faculty, interaction with industry experts

04 Data Repository and Access for AI Development
AICRA has developed data repository for learning purpose

05 Exposure to Industry and Investors
Students to get chance to showcase their projects at various platforms/competitions

ISL connects communities to address local challenges through innovative technologies. By fostering collaboration, we create prototypes and solutions that empower cities, driving sustainable development and enhancing quality of life.

01



Students at India STEM Lab would be identifying real-life problems as well as regional level challenges and can invent technology solutions to such problems.

Most common challenges which can be solved by technologies:

- Waste management
- Pollution
- Public Transportation
- Child & woman safety
- Water management
- Energy crisis
- Disabled Friendly Infrastructure
- Security Challenges

And many more..

Regional & Local Challenges

02

Build to Execute

INDUSTRY FOCUSED INNOVATION

AICRA focusing on major industry that boost Indian economy. We have also engaged experts from these industries as well as formed Technology Governance Steering Committees. Students at India STEM Lab would be making projects related to these industries (not limited)

Aerospace

03

Guidance By Industry Experts

TECHNOLOGY GOVERNANCE STEERING COMMITTEE (TGSC)

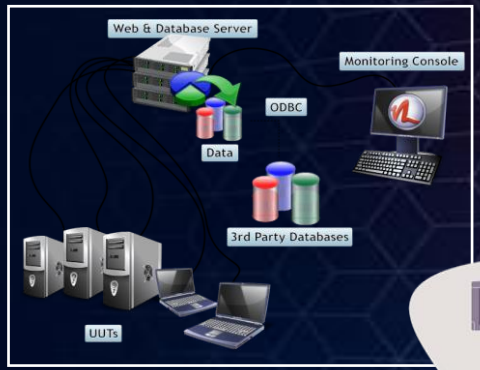
AICRA takes help of industry and experts to identify and design projects for the area of development in STEM Labs

- 3500+ Industry and academia members
- 1800+ Technical Consultants listed at GRAPES
- Various Govt. Departments' officials engagement at states/center
- 800+ Individual researchers and professionals

04

Empowering Insights Through Data Access

DATA REPOSITORY & ACCESS



AICRA has setup a strong database server which is accessible to Learners at India STEM Lab

- Students may save various category of data which can be used for AI based module development
- Student may also access data for industry based project development*

INTERNATIONAL PLATFORM TO SHOWCASE STEM EXCELLENCE

MAZE SOLVER | ROBO RACE | ROBO SOCCER | ROBO HOCKEY | RC CAR RACE | BOTSCOMBAT | WATER ROCKET | LINE FOLLOWER | SUMO BOTS | DRONE SOCCER | DRONE RACING | RC PLANE | DRONE RESCUE | INNOVATION

INSTITUTE RECOGNITION AND ENGAGEMENT IN VARIOUS NATIONAL/INTERNATIONAL PLATFORM

Empowering Futures with Certification

CERTIFICATION

Students participating in the India STEM Lab will receive certification from AICRA, validating their skills and knowledge in STEM disciplines. This certification enhances their educational credentials and prepares them for future career opportunities.

Global Reach, Local Impact, Everywhere

STUDENT DELEGATEION PROGRAM

The India STEM Lab student delegation program enables selected students to visit international institutions, gaining insights into global advancements in STEM education. In return, international students will visit India, fostering cross-cultural exchange and collaboration. This initiative aims to enhance learning experiences and inspire innovation among future leaders in technology and engineering.

Empowering Educators, Ensuring Excellence

FACULTY DEVELOPMENT PROGRAMS & ENGAGEMENT WITH REGULARITY

The faculty development program focuses on enhancing educators' skills in STEM teaching and integration. By engaging with government regulations, the program ensures compliance and alignment with educational standards. Continuous professional development workshops and seminars empower faculty to adopt innovative teaching methods, ultimately enriching the learning environment for students in STEM fields.

ISRO's Mission Chandrayaan 3 Team Telecast on DD

Dr. Jitendra Singh, Minister of Science & Technology, interacting with new start-ups

Sri Nitin Gadkari, meeting institute representative who is showcasing training facility development at his institute

Sri Rajesh Pathak, Secretary, Technology Development Board, DST evaluating projects of students

Inspiring Innovation, Fostering Entrepreneurship

INSPIRATION, MOTIVATION AND DEVELOPING STARTUP CULTURE

Connecting students and faculty with industry leaders through various programs inspires innovation and entrepreneurship. By fostering mentorship and collaboration, we motivate the educational community to explore startup culture, encouraging creative problem-solving and critical thinking. This dynamic environment cultivates future leaders equipped to thrive in the evolving landscape of technology and business.

Celebrating Excellence, Inspiring Achievement

REWARDS & RECOGNITION

AICRA organize many academia and industry oriented reward & recognition programs. These rewards and recognition program celebrates excellence in education by acknowledging outstanding institutes, teachers, and students.

Through awards and incentives, we motivate academic achievement, innovative teaching practices, and student engagement. This initiative fosters a culture of excellence, encouraging continuous improvement and inspiring all stakeholders to strive for success in their endeavors.

Filling up "Online Institute Membership" Form
Existing institute members doesn't need to fill up form. They simply need to apply for ISL

Student Members Enrolments for ISL
Post getting approval of ISL Grant, MoU to be signed off between AICRA and School. Further school would start enrolling students for membership.

Code Creation for Portal and Fee Collection
Dedicated portal would be created for school to manage all student's account. Fee would also be collected and deposited via portal.

Lab Installation
Post confirmation of student's count, budget would be allocated to setup ISL. Material to be dispatched and Trainers to be deputed.

Efficient Setup, Strategic Implementation, Timely Launch

IMPLEMENTATION TIMELINE

The India STEM Lab setup includes site selection, lab creative design, and equipment installation, followed by trainer's deployment and training schedule making. The timeline spans appx 30 days, concluding with a grand opening event to launch the store and attract initial customers.

Timelines	
School apply for AICRA Membership and submit the request form (Online/Offline)	Day 1
AICRA STEM Team select the school for grant and approve the lab	Day 4
School sign off the agreement/MoU	Day 7
Site reiki and scheduling (Virtual or via DC) , Panel Creation for Registration	Day 10
Student registration completion online, fee deposit in portal	Between Day 11 to 20
STEM Lab installation and briefing	Between Day 21 to 25
STEM Lab handover and Trainer Deployment, Making training schedule	Between Day 26 to 30

india STEM MISSION

AICRA
All India Council For Robotics & Automation

EMPOWERING DREAMS, FORGING FUTURES

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India STEM lab is a community of students in a particular education institute who are willing to acquire knowledge on Robotics. The club will be offered training programs mixes the creativity and curiosity of young students with the hands-on education aspects of Science & Technology. With focus on robot building and Science based activities, Our learning methodology takes up innovative and challenging ideas and converts them in to solutions in the form of working robots or engaging science experiments.